

Children's Evaluations of Disclosing and Omitting Information in Close Relationships

Start Date

September 2024

Research Question

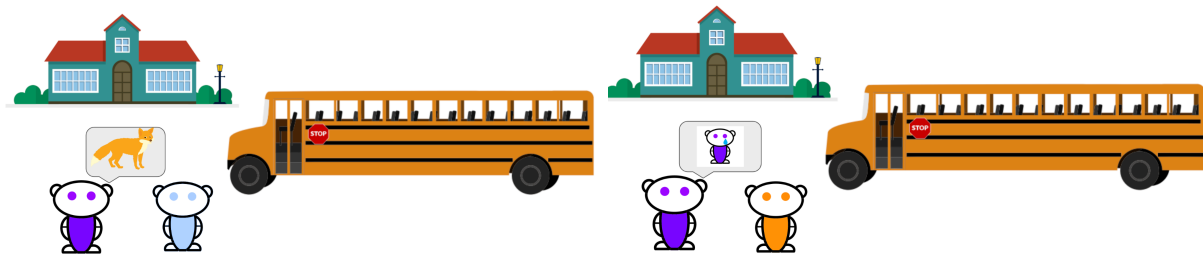
In the present study, we aim to investigate children's inference of withheld or omitted information in close relationships. In previous findings of children sharing sensitive information with close social partners and withholding from social partners who are not close (Anagnostaki et al., 2013), we are curious to see how children reason about withholding personal information in close relationships. We know that children as young as six years old can distinguish between different types of information (Anagnostaki et al., 2013). Likewise, children are also sensitive to the relationship to whom they are telling the information (Lieberman and Shaw, 2018). We will probe this kind of reasoning in three parts: determining who is closer based on information shared, determining which type of information to share to become closer, and determining what type of information should be shared in a close relationship.

Methods and Design

This is a within-subjects design and will be conducted fully online via Zoom. Each child will participate in three parts.

Part 1: How do children evaluate close relationships based on the type of information shared?

There will be two corresponding conditions. Children will hear vignettes about a protagonist who learns an animal fact in school and is also feeling an emotion. In one condition the protagonist will be feeling happy. This will be referred to as the happy condition. In the other condition the protagonist will be feeling sad. This will be called the sad condition. In each condition, the protagonist encounters two classmates individually. We tell children that the protagonist shared their emotional state with one classmate and did not share the fact, then shared the fact with the other classmate and did not share their emotional state. Here we are asking whether children think that people are closer if they share personal rather than factual information.

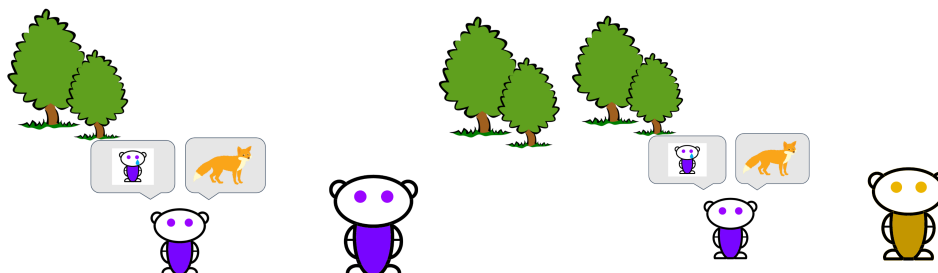


Part 2: What kind of information do children think is best to use to become close?

The second part of this study involves children learning that the same protagonist in the previous trials wants to become close friends with a new character. We then ask the participants to choose whether the protagonist should share the information about the fact or their emotional state. To reiterate, the emotional state would be happy in the happy condition and sad in the sad condition.



Part 3: What type of information do children expect to be shared in a close relationship?

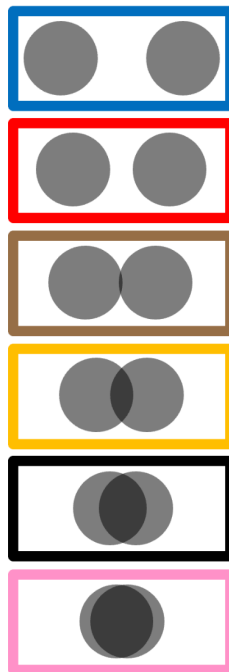


The final part of this study will test children's intuitions about what kinds of information should be shared in a close relationship. We will separately show the participants the

protagonist's mother and the protagonist's friend's mother who both want to know how the protagonist's day was at school. We will then ask children to choose if the protagonist should share the fact or their emotional state.

Warmup:

In this experiment, we utilize the scale from Aron et al.'s (1992) inclusion of others in the self as a measure of social closeness. We give the participants three warm-up questions so that they can become familiar with using the scale. We will ask the participants to select the box that best represents the relationships we ask them about. The three questions are as follows: mother and child, strangers, and friends.



Part 1 Test Trials

After we tell children the vignettes, we will ask them to recall which character was told the fact and which character was told the emotion. This will serve as an attention check. Then we will then ask children to rate how close each of the classmates are to the protagonist using the social closeness circle measure. This measure is adapted from representations of the self and others. We will present six circles that overlap each other at varying distances and ask the participants to choose which circle best represents the closeness of two characters in the trial. Next, we will ask children which classmate they think is a better friend, which classmate the protagonist would share ice cream with, and which classmate the protagonist would share candy with. These are binary forced-choice measures.

Part 2 Test Trials

We will then have an additional trial where the protagonist meets a new character who they wish to be close friends with, and we will ask children whether the protagonist should share the information about their emotional state or the fact to achieve this goal. This will be a binary forced-choice measure.

Part 3 Test Trials

Then there will be two more trials where the protagonist encounters their own mother and their friend's mother, who wants to know how school was that day. We will ask children whether the protagonist should share the fact or their emotional state with each character. This will be a binary forced-choice measure.

Participants

Children aged 6 years and 0 days to 9 years and 365 days of age. We chose this age range because it is consistent with past literature that suggests that children at this point in development have adequate theory-of-mind reasoning and social perspective-taking skills to navigate these tasks (e.g. Liberman and Shaw, 2018; Woo et. al., 2024).

Sample Size

We plan to test 56 children. This sample size is consistent with prior research that has found similar effects (e.g. Woo et. al, 2024).

Logic and Hypothesis

Part 1:

We hypothesize that children will think the classmate with whom the protagonist shares their emotional state will be closer, a better friend, and more likely to share ice cream and candy than the classmate who was told the fact. We believe that this hypothesis will hold across happy and sad conditions.

Part 2:

For the new friend trials, we hypothesize that in the sad condition, the children will think that the protagonist is better suited to share their emotional state than the fact. In the happy condition, we hypothesize that children will think that the protagonist should share the fact instead of their emotions. This is because children may think that feeling sad is more vulnerable to share with someone than feeling happy and will choose the sad emotion to share in order to become close. However, in the happy condition

children may find value in the fact to establish closeness, as it shows the protagonist has knowledge that the other character may be interested in learning about.

Part 3:

We hypothesize that children will think that the protagonist should share information about their emotional state with their mother across conditions. We also hypothesize that children will think that the protagonist should share information about the fact with the protagonist's friend's mother across conditions.

Primary and Secondary DVs

Part 1

The dependent variables are how close the children rate the protagonist to each of the characters (scale from 1 to 6).

Closeness will also be measured by a series of binary forced-choice items, which consist of the following:

- which classmate is a better friend to the protagonist
- which classmate the protagonist would share an ice cream cone with
- which classmate the protagonist would share candy with

Part 2

Closeness is measured by which type of information (emotion or fact) would be shared in order to become close friends with a new character.

Part 3

Closeness is measured by which type of information (emotion or fact) should be shared with a presumed close social partner (the protagonist's mother) and a presumed non-close social partner (the protagonist's friend's mother).

Primary Analyses

Part 1

Social Overlap:

To analyze the circles measure, we will use a Bayesian linear mixed-effects model.

Closeness ~ Condition (Trial + 1 | ID)

Where closeness is from 1 to 6.

Better Friend:

We will use a Bayesian binomial test to measure how often children choose the character that was told the fact or emotion as a better friend.

Saliva Sharing:

We will use a Bayesian binomial test to measure which character the children think the protagonist would share an ice cream cone with.

Food Sharing:

We will use a Bayesian binomial test to measure which character the children think the protagonist would share candy with.

Part 2

We will use a Bayesian binomial test to analyze whether children think that the protagonist should share the fact or emotion in order to become friends with a new character.

Part 3

We will also use a Bayesian binomial test to analyze whether children think that the protagonist should share the fact or their emotional state with their mother and friend's mother.

Inference Criteria

In all analyses, we will assess whether the value 0 falls within the 95% credible interval. The absence of 0 in the 95% credible interval will be interpreted as supporting evidence for the presence of effects, whereas the presence of 0 in the 95% credible interval will be interpreted as supporting evidence for the null.

Counterbalancing

We will counterbalance the order of conditions we will present to the participants as well as the order of the protagonist disclosing and omitting information.

Exclusion Criteria

We will exclude data if there is constant familial interference, procedural error, or equipment failure where the participant is unable to hear all of the vignettes.

Familial Interference:

We require that caregivers avoid intervening or guiding their children's behavior.

Caregiver interference includes caregivers doing something that might bias their child's

responses (e.g., speaking to their children about the events, telling their children the answers, etc.).

Because we are testing children over Zoom and they are in their family homes, we have less control over the testing environment. If siblings or others in the home environment consistently distract or interfere in a way that might bias a child's responses (as in the examples of caregiver interference above), we will have reason to exclude the participant on the basis of familial interference.

Procedural Error:

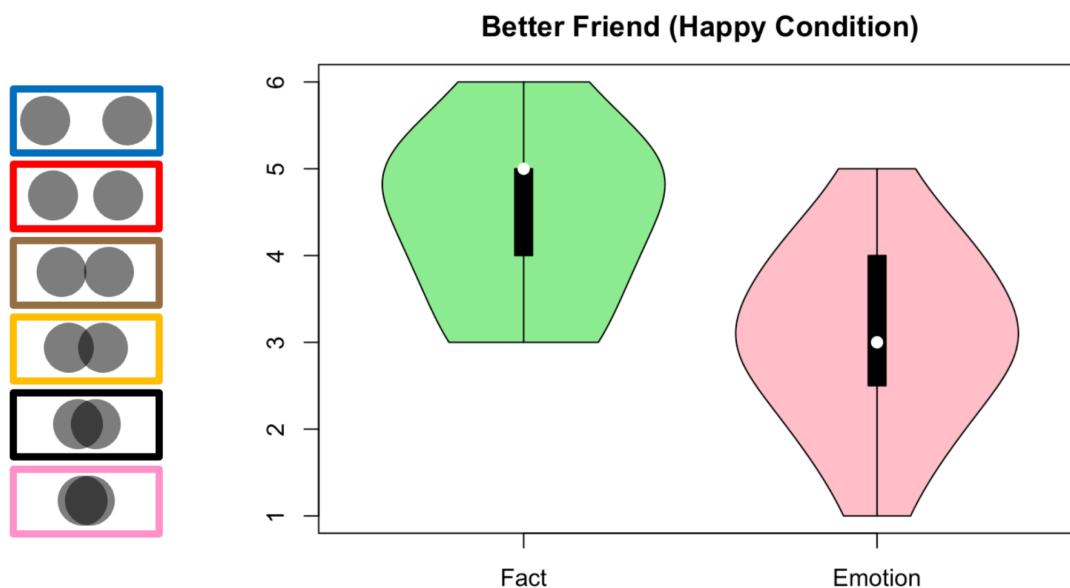
This includes deviation from the standard procedure of the study such that the participants do not see all vignettes, trials, etc.

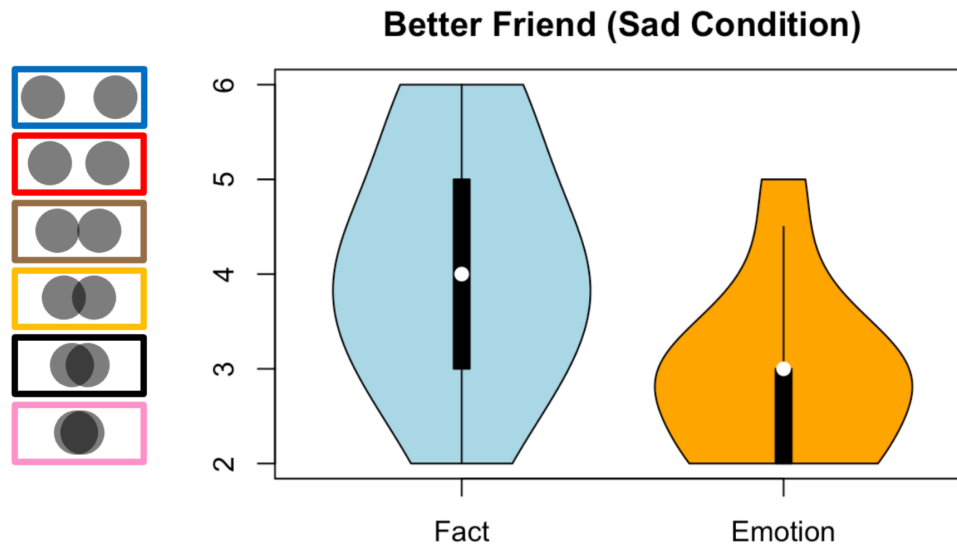
Equipment Failure:

This can occur when there is a lack of adequate internet access to complete the study or if the device the participant or experimenter is using malfunctions to the point where the study cannot be completed.

Pilot Data

We collected a pilot sample (N=15) of 6-9-year-old children to see if there were any preliminary findings for part 1 and part 2 of the study (part 3 was added after the pilot sample was collected). We saw a trend where children chose the classmate who was told the emotion as a better friend and was rated closer on the circle measure than the classmate who was told the fact. For part 2, we observed a trend where for the happy condition, children chose the fact to be shared in order to become close to the new character. Conversely, the results were roughly at chance for the sad condition.





Early Study Design

Before we came up with the current study outlined above, we first designed a study and collected a pilot sample of 6-9-year-old children ($N = 11$) using vignettes of a protagonist who learns about an animal fact in school and who also receives a bad grade on a test. We had three scenarios where the protagonist encountered a different type of relationship: a classmate, a best friend, and a stranger. For each relationship, the protagonist either told their social partner that they got a bad grade on the test and did not tell them about the animal fact or vice versa for a total of six trials.

We measured the moral evaluation that the participants gave for each scenario in regard to the protagonists' actions and also the emotional attribution for both the protagonist and the social partners. We hypothesized that the children will be sensitive to the type of information that is being withheld and will give more negative moral evaluations and emotional attributions when the protagonist withholds personal information (the bad grade), especially from a close relationship (the best friend) more than other relationships. However, we found that the children were not sensitive to the relationship in their evaluations and rated omitting the bad grade as morally bad across relationships and disclosing the fact as morally good across relationships. This led us to modify the design of the study.